

Locality Gates Dimensionality: When (and Why) a 2D Computational Medium Beats Its 1D Serialization

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Abstract

Does laying a problem out in two dimensions help a neural network reason about it? Prior work returns conflicting answers because it conflates three separate questions: whether 2D structure helps as an *input encoding*, as a *computational medium*, and as a *learned representation*. We disentangle them with a sequence of pre-registered experiments in which every model arm consumes the same lossless information and arms differ only in the connectivity of their operator. We find that *locality gates dimensionality*: a 2D layout confers a computational advantage if and only if the operator is constrained to be spatially local, and the task’s dependency structure has low bandwidth in 2D but high bandwidth under every 1D linearization. Under these conditions the advantage is large and robust: a 2D-local operator solves iterative relational tasks at twice the problem width of a parameter- and window-matched 1D-local operator (a gap that survives a space-filling-curve control), and a 2D-local rule trained on small grids transfers to unseen grids up to $11\times$ larger in area across three task families (connectivity, metric, dynamics) while its 1D counterpart drops to chance; all headline effects hold at 5 seeds under Holm correction ($p < 2 \times 10^{-5}$). A depth taxonomy sharpens the picture: transfer is *flat* when the task’s intrinsic computation depth is bounded, and *fades* only when required depth scales with problem size. Outside these conditions the advantage vanishes: global attention erases it in-distribution (while a curriculum-trained iterative Transformer that masters the training sizes still transfers nothing to larger ones — even on a bounded-depth task that its fixed iteration budget covers at every size), recurrent message passing on given graphs already size-generalizes without any layout (a boundary we reproduce in-house: a stabilized recurrent GNN reaches 1.00 at every width on given-graph reachability), and learned differentiable placements of implicit structure lose to a set-transformer. We compile these results into a decision procedure for when a 2D-local computational medium is the right inductive bias, and argue that its practical home is native-grid, locality-constrained computation rather than general relational reasoning.

1 Introduction

Two-dimensional layouts are ubiquitous in human reasoning artifacts — blueprints, circuit diagrams, board games, spreadsheets — and a classical argument holds that diagrams support reasoning *computationally*, by making related items adjacent so that inference requires only local search (Larkin and Simon, 1987). Modern deep learning offers a natural test bed for this hypothesis: does giving a neural model a 2D arrangement of a relational problem, rather than a 1D serialization of the same content, improve its reasoning?

The question is not academic: the field currently bets on both sides of it daily. Vision Transformers serialize images into 1D patch sequences, discard the convolutional locality prior, and win at scale, explicitly framing large-scale training as trumping inductive bias (Dosovitskiy et al., 2021); neural PDE surrogates and cellular automata bet on local 2D operators; and the default practice of the LLM era is to serialize grid-structured worlds — tables, boards, ARC-style puzzles — into token streams, where their difficulty is a live concern. Our own program produced both answers in consecutive pre-registered phases: a null for 2D structure under global attention, and a large positive under a locality constraint, reconciled only once the question was conditioned properly.

Empirically, then, the question is unsettled, and we argue this is because it is usually posed without controlling *what kind* of 2D usage is at stake:

1. **2D as input encoding.** Tokens carry (x,y) features; the operator is global attention. Here a set of positioned tokens is computationally just “a set with two extra features,” and permutation-invariant attention is already robust to the geometric transformations (e.g. translation) that 2D structure is supposed to buy. In our own earlier pilot (summarized in §7.1), three operationalizations of a 2D positional bias all failed to beat a plain set-attention baseline.
2. **2D as computational medium.** The state lives on a 2D field and the operator is *spatially local* (convolution, windowed attention, cellular automaton). Now the layout determines which interactions are cheap: locality makes dimensionality matter.
3. **2D as learned representation.** The model must *discover* a 2D placement of structure that is not natively spatial. This adds a hard joint optimization on top of (2).

These three regimes are not an ad-hoc list: they are cells of a 2×2 design spanned by two orthogonal axes, *operator connectivity* (global vs. local) and *layout provenance* (given vs. learned). The fourth cell — a learned layout consumed by a global operator, familiar as learned positional embeddings in Transformers — is degenerate under this paper’s headline result: without locality, position is a feature rather than a schedule, and the cell inherits the verdict of regime (1). A scope note: throughout, we ask whether a 2D layout helps the model *compute*; uses of 2D as an output or interface representation for humans (diagrams, visualizations) are a separate question and out of scope.

This paper’s contribution is a controlled, pre-registered characterization of regime (2), together with boundary experiments delimiting it against (1) and (3). Our headline finding is conditional: the 2D advantage exists, is large, and is mechanistically explainable, but only inside a regime whose boundaries we can state precisely. We map that regime; we do not advocate beyond it.

Concretely, we contribute:

- **Mechanism I — bandwidth separation (§4).** On a $W \times W$ grid, the dependency graph of a local iterative computation has bandwidth 1 under the 2D layout but bandwidth $\geq \min(W, H)$ under every 1D linearization (Chvátalová, 1975). At matched parameters, window size, and depth, a 2D-local operator clears a reachability task at threshold-width $W^* = 8$ versus $W^* = 4$ for its 1D-local counterpart — and a space-filling-curve (Hilbert-style) serialization does not close the gap ($W^* = 4$), turning the graph-bandwidth theorem into an observed floor.
- **Mechanism II — scale-equivariance and length-generalization (§5).** A weight-shared 2D-local rule is the *same computation* at every grid size, whereas a 1D serialization couples the learned kernel to the width W (a vertical neighbor sits W positions away). Trained on $W \leq 6$ and tested up to $W = 20$, 2D-local operators retain 0.73–0.96 accuracy across three task families while the matched 1D operator is at chance on every unseen size.

- **Mechanism III — a depth taxonomy (§6).** Whether transfer is flat or fading is predicted by whether the task’s intrinsic computation depth is bounded (local dynamics, fixed-radius metric queries: flat at 0.89–0.96) or scales with problem size (reachability: fades under a fixed budget). An adaptive-depth neural cellular automaton does not rescue the scaling case; the depth bound is a property of the task class rather than a fixable artifact.
- **Boundary map (§7).** Three pre-registered negative results delimit the regime: (i) removing the locality constraint (global attention) erases the advantage; (ii) when the sparse graph is *given*, recurrent message passing already size-generalizes up to $1000\times$ without any spatial layout (Grötschla et al., 2022) — no room for a 2D win; (iii) when the layout must be *learned* from implicit structure, a differentiable Sinkhorn placement trains but underfits, and a set-transformer beats it.
- **A decision procedure (§8)** that a practitioner can apply to predict whether a 2D-local medium will help on their task.

A methodological note: every gate in this program was pre-registered (win/null thresholds committed to version control before the corresponding run), with matched-information and matched-budget constraints (§3.3), and with purpose-built controls: a space-filling-curve serialization, a 1D-intrinsic task, a GNN positive control, and a random-layout control. Several of the controls changed the conclusions (§4.1, §5.1, §7.3); we report the null and inconclusive outcomes alongside the positives.

What this paper does not claim. Our results do not support the reading that “2D representations reason better than Transformers”: under global attention the layout is irrelevant (§7.1), and nothing here bears on large-scale language models. Nor do we claim that 2D-local operators beat graph networks in general: on given graphs, stabilized recurrent GNNs size-generalize without any layout (§7.2), and our own GNN arm is a plain fixed-depth message-passing network used as a learnability control rather than a representative of the strongest graph methods (§5, §5.2); when edges must instead be *estimated* from features, the estimated graph itself degrades with size (§7.3). The claims are exactly the conditional ones: among matched local operators, on tasks with the stated bandwidth structure, the 2D layout wins, transfers across scale, and does so for stateable reasons.

2 Related work

Size- and length-generalization. GNNs trained on small graphs often fail on larger ones (Yehudai et al., 2021); the number of message-passing rounds bounds what graph problems are solvable at all (Loukas, 2020). Recurrent architectures with stabilization extrapolate dramatically better: Schwarzschild et al. (2021) on mazes and prefix sums with recurrent CNNs, IterGNN (Tang et al., 2020), and most relevantly Grötschla et al. (2022), whose recurrent GNNs (skip-to-input, L_2 state regularization, edge convolutions) extrapolate to graphs $1000\times$ larger on path, prefix-sum, and distance tasks. Our §7.2 treats this as a boundary condition: on *given* sparse graphs the size-generalization problem is substantially solved without any spatial layout. Benchmarks such as CLRS-30 (Veličković et al., 2022) formalize the out-of-distribution axis; Delétang et al. (2023) do the same for sequence models; see also Xu et al. (2021) on extrapolation in feedforward versus graph networks.

Oversquashing and rewiring. Long-range information on bottlenecked graphs is compressed through narrow cuts (Alon and Yahav, 2021); a substantial line of work rewires graphs to mitigate this (Topping et al., 2022; Black et al., 2023; Di Giovanni et al., 2023; Barbero et al., 2023). Our grids are deliberately

bottleneck-free; we do not compete with rewiring but note (§7.3) that “re-embed into a uniform lattice” as a *learned* rewiring did not survive its cheap gate.

Neural cellular automata. Weight-shared local 2D updates trained by gradient descent (Mordvintsev et al., 2020) have been extended toward computation (Bèna et al., 2025) and grid reasoning benchmarks (Xu and Miikkulainen, 2025); differentiable spatial computers with local read/write heads (Malhotra and Seghouani, 2025) are a recent addition to the same family. Our Mechanism II/III results give a controlled account of *why* such models length-generalize on grid tasks (scale-equivariance) and where they will not (depth-scaling tasks; our adaptive-depth null in §6.1).

Serialization of 2D structure. Two applied literatures have independently observed that flattening 2D data into a 1D sequence is not a neutral choice. Vision state-space models apply a 1D-local selective scan to serialized patches; there, scan order measurably affects accuracy (Hardan et al., 2025), and the field has progressed from raster scans through multi-directional to input-adaptive scan paths precisely because fixed orders disrupt spatial adjacency (Li et al., 2025). For language models, Lo et al. (2026) document “serialization friction”: on matrix tasks (transpose, Game of Life, LU decomposition), input rendered in native 2D layout outperforms 1D text serialization at the system level, with the gap widening in task size. These works establish the phenomenon in applied, system-level settings; what this paper adds is the controlled mechanism behind it — an information- and parameter-matched isolation of operator locality, a bandwidth account of why *no* fixed serialization can close the gap (the space-filling-curve control of §4.1 bounds the best case of the static scan-order design space), and the boundary conditions under which the effect disappears. The scan-order literature’s own trajectory — toward multi-path and adaptive scans that re-import 2D adjacency — is what the bandwidth account predicts. Conversely, Kirchmeyer and Deng (2023) show that ImageNet ConvNets built entirely from oriented 1D kernels match 2D convolutions; this is consistent with our analysis rather than contrary to it, because their oriented kernels still index the 2D layout (the medium stays two-dimensional; only the kernel is factorized), whereas the serialization our bandwidth argument penalizes is of the layout itself.

Diagrams and cognition. Larkin and Simon (1987) argued diagrams substitute cheap perceptual locality for expensive search. Our results are a machine counterpart: the benefit exists exactly when the architecture is forced to pay for non-locality.

Set versus sequence. Permutation-invariant set architectures are the established prior for order-irrelevant data (Zaheer et al., 2017; Lee et al., 2019). Our earlier pilot (§7.1) measured this gap directly on our own relational tasks and found *no additional* gain from 2D positional structure under global attention — the observation that motivated the present locality-centered design.

3 Experimental framework

3.1 Substrate

All tasks in the main program live on a $W \times W$ grid of cells. Each instance is presented as a 6-channel field: wall indicator, source marker, target marker, a scalar value channel, and normalized x, y coordinates. The information content is identical for every arm (constraint **H1**): sequence arms receive the same channels in a fixed serialization order, and the coordinates are explicit features, so no arm sees more than another; arms differ only in operator connectivity.

3.2 Arms

- **2D-local**: a depth- L stack of 3×3 convolutions (weight-shared variants in §5–6), hidden width 32.
- **1D-local**: kernel-9 1D convolutions over the row-major serialization — *identical parameter count and identical 9-cell window per layer* as the 2D arm (constraint **H2**); only the layout dimensionality differs.
- **1D-local (space-filling)**: the same operator on a generalized-Hilbert (locality-preserving) order — the strongest available 1D serialization.
- **global**: full self-attention (shallow, and separately a 16-iteration weight-shared Universal Transformer with pre-LN, per-step embeddings, gradient clipping, and parameter parity).
- **GNN**: message passing on the true 4-neighbor adjacency — handed the graph; a *plain fixed-depth* MPNN used as a positive control for task learnability (for the strongest graph methods, see §7.2 and the stabilized recurrent arm in §5.2).

3.3 Protocol

Every stage fixed its win/null thresholds in a pre-registration committed before the run, with a no-escape-hatch rule: a null closes the stage (no post-hoc threshold changes, no tuning toward a pass). Models are small (hidden 32), trained with Adam on balanced binary tasks. The headline positive comparisons (§4–5) were re-run at **5 seeds** with per-seed records; their effects are tested with one-sided paired t -tests (H1: 2D-local > 1D-local, paired by seed) under **Holm correction** across the five headline tests, and the original pre-registered gate criteria were re-checked unchanged on the 5-seed means (none flipped). The one-sided direction was fixed by the pre-registration, and the conclusions do not depend on it: under two-sided tests every p doubles and all five effects remain $< 4 \times 10^{-5}$ after Holm; moreover all five seeds agree in direction on every headline comparison, so even a distribution-free reading (sign test, its floor at $n = 5$) supports the same verdicts without the t -distribution’s normality assumption. Diagnostic and boundary/null results (§6.1, §7) retain their original 3-seed runs and are flagged as such — a null closed under the no-escape-hatch rule is not re-litigated with more seeds.

3.4 Tasks

- **F1 — reachability** (connectivity): random walls at density 0.28; source in the top row, target in the bottom row (forcing traversal along the high-1D-bandwidth axis); label = target reachable from source (4-connected). Classes are 50/50 balanced *and exactly wall-count matched*, so density heuristics carry no signal.
- **F3 — 1D-intrinsic control**: parity of a row prefix — the dependency runs along a single axis, so a 2D layout should give *no* advantage. Any 2D win here would indict generic capacity rather than bandwidth.
- **F4b — bounded metric query**: is the target within $K = 6$ hops of the source (target placed within a fixed radius 8)? The computation is identical at every W (scale-invariant by construction; see §5.1 for the failed first version and what the control caught).
- **F5 — local dynamics**: predict a queried cell’s state after $T = 2$ steps of a fixed totalistic cellular-automaton rule; the answer depends on a 5×5 neighborhood — local in 2D, dispersed across $\sim 2TW$ indices in any row-major order.

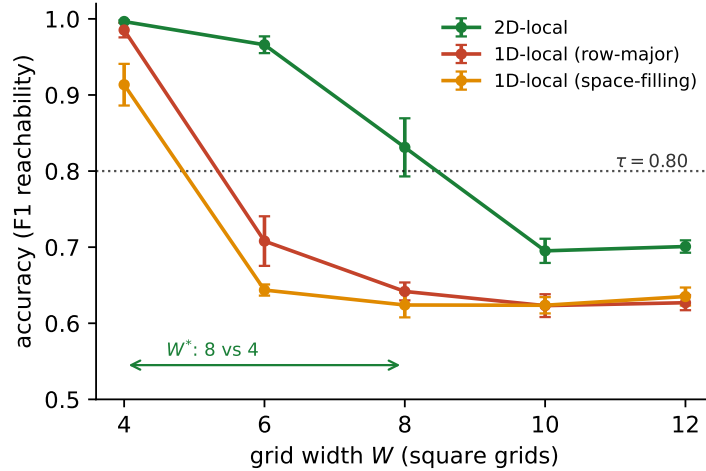


Figure 1: **Bandwidth separation at fixed size** (F1 reachability; 5 seeds, mean $\pm$ s.d.). At matched parameters, window size, and depth, the 2D-local operator clears the $\tau = 0.80$ threshold up to $W^* = 8$; both 1D serializations — row-major *and* the locality-preserving space-filling curve — cap at $W^* = 4$, as the grid-bandwidth theorem predicts.

4 Mechanism I: bandwidth separation at fixed size

The first mechanism is a bandwidth separation. For an iterative local computation on a $W \times H$ grid, the 2D layout has dependency-graph bandwidth 1 (all one-step interactions are lattice-adjacent), while *every* 1D linearization has bandwidth $\geq \min(W, H)$ (Chvátalová, 1975). A local operator with a fixed window must therefore spend $\sim \min(W, H)/\text{window}$ steps to move information one grid row under any serialization; the 2D layout needs one.

We measure this with the threshold-crossing width W^* : the largest W (square grids) at which an arm’s mean accuracy on F1 stays $\geq \tau = 0.80$, at matched depth 16 and matched 9-cell windows. The pre-registered pass criterion was $W^*(2D) - W^*(1D) \geq 2$.

The outcome, at 5 seeds (Figure 1; per-seed W^* in brackets):

W	4	6	8	10	12	W^* (per-seed)
2D-local	1.00	0.97	0.83	0.70	0.70	8 [8,8,8,8,6]
1D-local (row-major)	0.99	0.71	0.64	0.62	0.63	4 [4,4,4,4,4]
1D-local (space-filling)	0.91	0.64	0.62	0.62	0.64	4 [4,4,4,4,4]

Table 1: Fixed-size threshold sweep on F1 (mean accuracy over 5 seeds).

The gap (+4) is wide at the crossing ($\Delta \approx 0.19\text{--}0.26$ at $W = 6\text{--}8$), i.e. not a threshold artifact, and is seed-stable: both 1D arms sit at $W^* = 4$ in **every** seed. The per-seed mean-accuracy advantage over $W \in \{6, 8, 10, 12\}$ is $+0.148 \pm 0.012$ versus row-major (paired $t = 27.6$, $p_{\text{Holm}} = 1.2 \times 10^{-5}$) and $+0.167 \pm 0.013$ versus the space-filling order ($t = 29.2$, $p_{\text{Holm}} = 1.2 \times 10^{-5}$).

A note on what the theorem does and does not entail. Given a fixed window, the *direction* of this separation is entailed by the bandwidth bound; what this section contributes is the measurement under controls. What the bound does *not* entail: that gradient-trained models realize the floor cleanly (expres-

sivity does not imply trainability — our global baselines are expressive enough for these tasks yet fail to train, §7.1), the magnitude and seed-stability of the gap, and the transfer phenomena of §5–6, which are not corollaries of the bound.

4.1 The serialization objection, defeated empirically

The obvious objection — “row-major is just a bad order” — is what the space-filling-curve arm answers: the locality-preserving Hilbert-style order, whose consecutive cells are always grid-adjacent, performs *no better* than row-major ($W^* = 4$, and slightly worse in the working range). This is the bandwidth theorem made empirical: preserving locality *along* the curve cannot fix the $\Theta(\min(W, H))$ reach needed *across* it. No choice of serialization closes the gap.

4.2 The capacity objection, defeated by the F3 control

On the 1D-intrinsic control task the ordering reverses: 1D-local reaches $W^* = 10$ versus 8 for 2D-local (5 seeds; mean 2D–1D gap over the sweep -0.037 , two-sided $p = 0.34$ — no 2D advantage, direction negative). The 2D arm is not generically stronger; it wins exactly where the bandwidth analysis says it should, and loses where the task’s dependency structure favors the serialization.

5 Mechanism II: scale-equivariance and length-generalization

The second mechanism is scale-equivariance. A weight-shared 2D-local rule commutes with changes of grid size: the learned kernel implements the same one-step interaction on any $W \times H$ field. A 1D serialization breaks this: the distance between vertically adjacent cells is W , so a kernel calibrated at training width is *semantically miscalibrated* at any other width. The prediction is that 2D-local rules trained small transfer to larger unseen sizes, while 1D-local rules do not merely degrade; they transfer nothing.

To test transfer, we train on mixed $W \in \{4, 5, 6\}$ and evaluate on unseen $W \in \{7, \dots, 20\}$ (up to $11 \times$ the training area). The pre-registered pass criteria per family: mean 2D–1D gap ≥ 0.10 over $W \leq 10$ –12, $2D \geq 1D$ everywhere, and the GNN positive control ≥ 0.70 at $W = 8$ (establishing the task *is* length-generalizable by a scale-free learner).

On reachability (F1), the transfer picture is stark (5 seeds; all test W unseen; Figure 2, left):

W	8	10	12	16	20
2D-local	0.83	0.75	0.73	0.73	0.73
1D-local (row-major)	0.51	0.51	0.52	0.51	0.49
GNN (positive control)	0.80	0.58	0.51	0.51	0.51

Table 2: Length-generalization on F1 (mean accuracy over 5 seeds; train $W \in \{4, 5, 6\}$).

The 1D arm is at *chance* on every unseen width — the strongest single contrast in this program; the space-filling serialization also fails to transfer (0.57–0.63 across unseen $W = 7$ –12 in a 3-seed diagnostic run). Notably, the plain fixed-depth GNN handed the true adjacency generalizes one step out (0.80 at $W = 8$, satisfying the positive control) and then **collapses** (0.58 at $W = 10$, chance by 12), while 2D-local holds 0.73 out to $W = 20$. We stress what this does and does not show: the collapse indicts *plain fixed-depth* message passing only, and it is a stabilization failure rather than anything intrinsic to graph computation — adding the standard stabilizers (input skip, GRU state update, L_2 state regularization)

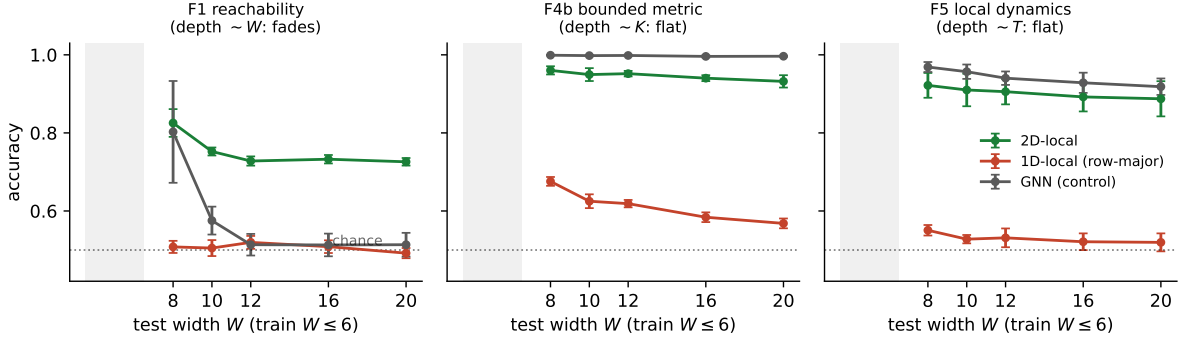


Figure 2: **Length-generalization across the reasoning suite** (train $W \leq 6$, test on unseen W ; 5 seeds, mean $\pm$ s.d.). The 1D-local arm transfers nothing (chance-to-weak on every family), while 2D-local transfers on all three — flat where intrinsic depth is bounded (F4b, F5) and fading where required depth scales with W (F1). On F1 the plain fixed-depth GNN handed the true adjacency satisfies the positive control at $W = 8$ and then collapses (§5.2 evaluates a stabilized recurrent GNN), while 2D-local holds 0.73 out to $W = 20$.

repairs it completely (§5.2); a stabilized recurrent GNN in the style of Grötschla et al. (2022) is evaluated directly there, and given graphs remain the territory of §7.2.

Across the full suite (train $W \leq 6$, test $W = 8 \rightarrow 20$, 5 seeds; gap = per-seed mean of 2D–1D over the pre-registered range $W \leq 12$), the same pattern holds in all three families:

Family (computation type)	2D-local	1D-local	gap (mean $\pm$ sd; p_{Holm})	GNN @8
F1 reachability (connectivity)	0.83 $\rightarrow$ 0.73	≈ 0.51	$+0.258 \pm 0.017$; 9.4×10^{-6}	0.80
F4b bounded metric ($K = 6$)	0.96 $\rightarrow$ 0.93	0.57–0.68	$+0.314 \pm 0.013$; 1.8×10^{-6}	1.00
F5 local dynamics ($T = 2$)	0.92 $\rightarrow$ 0.89	≈ 0.53	$+0.376 \pm 0.034$; 1.2×10^{-5}	0.97

Table 3: The three-family length-generalization suite (5 seeds).

Length-generalization of the 2D-local medium is therefore a *general* property across connectivity, metric, and dynamics computation, not a quirk of reachability. A further note on placement: the adversarial source/target geometry (top row $\rightarrow$ bottom row) exists *only* in F1, where it deliberately exercises the bandwidth mechanism; F4b and F5 place their queries direction-neutrally at random, and show the *larger* gaps ($+0.314$ and $+0.376$ versus F1’s $+0.258$). The advantage does not depend on rigging the geometry, and §5.2 confirms this on F1 itself.

5.1 What the positive control caught

Our first metric family used a width-coupled threshold ($K = 2(W-1)$). The GNN control *failed* on it (0.56), flagging that the target function itself was not scale-invariant — even a scale-free learner cannot transfer a size-dependent definition. Redesigning with fixed K (F4b) raised the control to 1.00 and the 2D arm to 0.93–0.96. We report this because it is easy to mistake such a confound for a model failure; the control isolates task-side from model-side non-transfer, and we recommend it as standard practice in length-generalization studies.

5.2 Robustness: neutral placement and a stronger graph baseline

Two checks respond to natural objections to the F1 design (5 and 3 seeds respectively; raw per-seed data in `robustness.json`).

Placement-neutral F1. F1’s top-row/bottom-row placement deliberately forces traversal along the high-1D-bandwidth axis; does the advantage require this rigging? We re-ran the length-generalization protocol on an F1 variant with source and target placed uniformly at random (same 50/50 balance and exact wall-count matching). The gap survives decisively: 2D-local $0.91 \rightarrow 0.64$ versus 1D-local $0.59 \rightarrow 0.52$ across unseen $W = 8\text{--}20$; per-seed mean gap $+0.212 \pm 0.010$ (paired $t = 47.0$, $p = 6.2 \times 10^{-7}$, 5 seeds), versus $+0.258$ under the adversarial placement. The forcing sharpens the contrast; it does not create it.

A stabilized recurrent GNN. Is the §5 GNN collapse representative of graph methods? No — and confirming this with our own data closes the loop on §7.2. We implemented a stabilized recurrent GNN following Grötschla et al. (2022): one weight-shared round with an input skip, an edge MLP, a GRU state update, and L_2 state regularization; variable training rounds and $3W$ rounds at inference; trained with roughly twice the epoch budget of the other arms, in the same fair-shot spirit as the Universal Transformer of §7.1. On the original given-graph F1 it size-generalizes *perfectly* — **1.00 at every test width up to $W = 20$** (3 seeds) — outperforming every other arm, including 2D-local. This is the §7.2 boundary reproduced in-house: *when the dependency graph is handed to the model, a stabilized recurrent GNN is the right tool and spatial layout adds nothing*. The 2D-local medium’s case is the complementary regime: locality imposed and structure spatial (positions and fields) rather than given as an edge list. (The arms also differ in what they must extract: the recurrent GNN receives wall-aware adjacency, while the local operators must infer connectivity from the raw field — so this comparison bounds the value of *being given the graph*, not architecture alone.)

6 Mechanism III: a depth taxonomy

The suite reveals systematic structure in *how* transfer behaves:

- **Bounded intrinsic depth $\Rightarrow$ flat transfer.** F4b needs $\sim K$ propagation steps and F5 needs T steps — independent of W . Both transfer nearly flat (0.93–0.96 and 0.89–0.92 out to $W = 20$).
- **Size-scaling depth $\Rightarrow$ fading transfer.** F1’s required depth grows like the path length ($\sim W$); under a fixed 16-layer budget accuracy fades ($0.83 \rightarrow 0.73$) as instances outgrow the budget.

6.1 Adaptive depth does not (yet) break the bound

The obvious fix — iterate a weight-shared local update $T \propto W$ times (a stabilized neural CA with variable-step training) — was pre-registered and returned **null**: the adaptive-depth CA beats the fixed-depth stack in the mid-range ($+0.11$ to $+0.13$ at $W = 10\text{--}12$) but ties it at $W = 16$ (0.72 vs 0.73) and *loses* at $W = 20$ (0.64 vs 0.72), consistent with drift accumulating over ~ 60 iterations. Meanwhile the fixed-depth 2D stack is itself a surprisingly strong flat generalizer (0.72–0.73 across $W = 8\text{--}20$). We record this as a real, unbroken limitation: at this scale, the 2D-local medium’s clean transfer story is for bounded-depth computation; depth-scaling tasks remain budget-limited.

Practical reading. Given a task, ask how its required propagation depth scales with instance size. Bounded $\Rightarrow$ a small 2D-local model trained on small instances will transfer essentially flat. Scaling $\Rightarrow$ expect graceful fading at fixed budget, and do not expect a naive adaptive-depth CA to fix it.

7 Boundary conditions: where the advantage does not exist

The regime of §4–6 is bounded by three negative results, each pre-registered.

7.1 Remove locality and the advantage vanishes

In an earlier pilot study we tested 2D structure as an *input encoding* under global attention: absolute Fourier positional features, relative-position attention biases, and a translation-invariant mean-centering model, on node-link relational tasks. All three converged on **2D-versus-set** ≈ 0 (e.g. $-0.013 / +0.012 / -0.058$ across task families under translation shift): set attention is already robust where 2D structure was supposed to help, and the only large effect was the known set $\gg$ sequence gap. Within the present program we made three attempts at a capable iterative-global baseline on F1. A $16\times$ -looped transformer failed to train at all (chance at every width), and a parameter-matched stabilized Universal Transformer reached only 0.70 even at trivial $W = 4$. The third attempt succeeded: with a width curriculum ($W = 4 \rightarrow \{4, 5\} \rightarrow \{4, 5, 6\}$), learning-rate warmup, and hidden width 64, the same Universal Transformer trains to 0.84 mean accuracy on $W \in \{4, 5, 6\}$ (3 seeds), and then transfers *nothing*: 0.68 at $W = 8$, 0.55 at $W = 12$, and chance (0.52–0.53) at $W \in \{16, 20\}$, while the 2D-local arm trained on the same widths holds 0.73 at $W = 20$ (Table 2). The boundary therefore no longer rests on a training failure: at this scale, a *trained* iterative-global model matches the local arms in distribution — where, per the input-encoding pilot above, layout is irrelevant — and cedes the entire length-generalization axis to the 2D-local medium. Global attention erases the layout advantage *at the sizes it is trained on*; it does not inherit the scale-transfer that locality provides. (Caveats: one recipe; larger-scale global models remain untested, §9.)

One confound remains in the F1 comparison: F1’s required depth scales with W , while the UT’s iteration count is architecturally fixed at 16 by its per-step embeddings, so its large- W failure could partly reflect depth budget rather than the operator. We therefore re-ran the identical recipe on the bounded-depth families, where 16 iterations suffice at every evaluated width (F4b needs $\sim K = 6$ propagation steps; F5 needs $T = 2$), under pre-registered thresholds (3 seeds; raw data in `a08b_global_bounded.json`). On F4b the UT trains to 0.96 at the training widths — above its F1 level — and still transfers nothing: 0.86 at $W = 8$, 0.69 at $W = 12$, and chance (0.50–0.54) at $W \in \{16, 20\}$, against the 2D-local arm’s flat 0.93 at $W = 20$. Depth budget is thereby eliminated as the explanation on this family: a trained global model with ample iteration depth still cedes the transfer axis, isolating the cause in the operator’s lack of scale-transferable (local, weight-shared) structure. On F5 the same recipe stopped just short of the pre-registered training gate (0.78 against the 0.80 bar), so per protocol it yields no transfer verdict; we report it as a training failure (for completeness, its unseen-size accuracies show the same decay to chance).

A note on capacity asymmetry. The trained Universal Transformer uses hidden width 64 — twice the hidden width 32 of the local arms (§3.2) — because that capacity was needed to get the global arm to train at all; the smaller, parameter-matched variant of our second attempt did not reach a usable training accuracy. The asymmetry therefore runs in the direction that favors the global arm, and its consequences cut one way: the UT’s in-distribution parity is achieved with a larger model than the 2D-local arm it is compared against, and its transfer failure cannot be attributed to capacity starvation. The comparison, if anything, flatters the global arm; the transfer gap survives it.

7.2 Given the graph, layout is unnecessary

Grötschla et al. (2022) show recurrent GNNs with skip connections, state regularization, and edge convolutions extrapolate path, prefix-sum, and distance tasks to graphs $1000\times$ training size. We treat this

as a boundary rather than a competitor: **when the sparse dependency graph is handed to the model, size-generalizing iterative computation is achievable with no spatial layout at all.** Our §5.2 arm reproduces this in-house: a stabilized recurrent GNN reaches 1.00 at every test width on given-graph F1. A 2D re-embedding can only add value when the graph is *not* given — which motivated the final experiment.

7.3 Learning the layout loses to a set-transformer

We built the implicit-structure case: a latent $W \times W$ grid with a smooth land/water field; input is an *unordered set* of cells with affinely scrambled coordinates (rotation + shear + noise; recoverable but not given), no adjacency list; the label is whether two marked cells share a 4-connected land component. The layout arm encodes the set, predicts per-cell positions, soft-assigns cells to lattice sites via Sinkhorn (Cuturi, 2013; Mena et al., 2018), and runs the §5 local-CA — trained end-to-end (gradients flow through the placement; verified). Controls: a set-transformer, a kNN-graph recurrent GNN, and a *random-layout* arm (same lattice + CA, placement not learned).

Result (train $W \in \{6, 8\}$, test $W \in \{12, 16, 20\}$; re-run at **5 seeds** with a $1.5\times$ epoch budget for this paper — the original pre-registered gate used 3 seeds and reached the same verdict): the learned layout extracts real structure — it beats random layout by $+0.11$ at $W = 20$ and transfers flat ($0.65 \rightarrow 0.62$) — but it *underfits* (0.65 at train size vs $0.70\text{--}0.71$ for the baselines), and the set-transformer is flat **and higher** (0.70 everywhere; -0.08 ahead at $W = 20$). Notably, the extra budget helped the baselines (set-transformer $0.67 \rightarrow 0.70$ versus the 3-seed run) but not the layout arm — the underfit is specific to the joint placement+execution objective, not a global training-budget artifact. Two of three pre-registered criteria fail $\Rightarrow$ null. One secondary observation survives: the kNN-graph arm *degrades* at scale ($0.71\text{--}0.73 \rightarrow 0.63\text{--}0.65$) while lattice and attention stay flat — regular structure transfers, data-dependent graphs degrade.

Boundary statement. The joint “infer the layout, then exploit it locally” problem is where the 2D-medium program currently stops: the placement is trainable but not yet competitive with a strong set baseline. This is the third independent reproduction (after the notation-utility and input-encoding pilots) of the pattern that permutation-invariant attention is the baseline to beat, and it usually is not beaten.

8 A decision procedure

For a practitioner asking “should I use a 2D-local medium for this task?” (Figure 3). Two notes on evidential status: the procedure synthesizes our controlled results, and its steps rest on evidence of differing strength — step 1 on a trained-global comparison at pilot scale (§7.1) and step 3 on a 5-seed null (§7.3); and two of its branch recommendations have been checked out-of-sample on external benchmarks (§8.1).

1. **Is your compute constrained to local operators** (hardware, latency, $O(N)$ -per-step budgets, or an architectural choice like CA/conv)? If no — if global attention is affordable and trainable on your task — the layout will not matter in-distribution (§7.1). If train-small $\rightarrow$ run-large transfer is itself the requirement, however, note that even our *trained* global ceded that axis (§7.1); locality then re-enters as the route to transfer.
2. **Is the dependency structure given explicitly** (edge lists)? If yes, use a recurrent GNN with stabilization; layout adds nothing (§7.2).

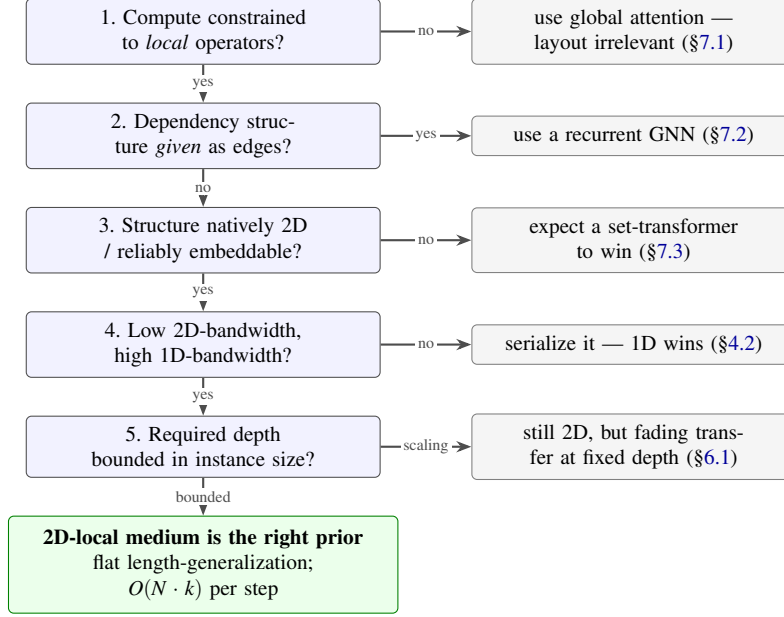


Figure 3: **The decision procedure** distilled from the mechanism and boundary results: five questions predicting whether a 2D-local computational medium will help on a given task.

3. **Is the structure natively 2D or reliably embeddable** (grids, images, boards, physical fields, spatial programs)? If it must be *learned* from implicit structure, expect a set-transformer to win at current recipes (§7.3).
4. **Is the dependency bandwidth low in 2D but high in 1D** (interactions spatially adjacent, no serialization keeps them adjacent)? If the task is 1D-intrinsic, serialize it (§4.2).
5. **Does required depth scale with instance size?** Bounded $\Rightarrow$ expect flat length-generalization from a small trained model (§6). Scaling $\Rightarrow$ expect fading at fixed depth; adaptive-depth CAs do not currently fix this (§6.1).

Yes to 1, 3, 4 (and ideally “bounded” at 5) is the regime where the 2D-local medium is, in our data, decisively the right prior — with per-step cost $O(N \cdot k)$ versus $O(N^2)$ for attention, and training possible entirely on small instances. Natural application domains are native-grid computation under locality constraints: spatial accelerators and systolic-style hardware mappings, cellular/field simulations, and grid-based reasoning benchmarks.

8.1 Out-of-sample checks of the procedure

The procedure synthesizes our own controlled experiments; a fair objection is that it had never been tested on data we did not design. We checked two of its branch recommendations on external benchmarks.

Given-graph branch (step 2) — official CLRS-30 BFS. Using the benchmark’s own sampler (dm-clrs 2.0.3; Veličković et al., 2022), we trained at the standard size $n = 16$ and evaluated pointer accuracy at $n \in \{16, 32, 64\}$ (3 seeds). The step-2 recommendation — use a stabilized recurrent GNN; no layout — holds exactly: the §5.2-style RecGNN reaches 1.00/1.00/0.95, while a plain fixed-depth MPNN degrades (0.98/0.91/0.58) — reproducing the in-house plain-versus-stabilized contrast on official benchmark data.

Native-2D branch (steps 3–5) — Schwarzschild et al. mazes. On the easy-to-hard maze benchmark (Schwarzschild et al., 2021) (predict the optimal-path mask; the metric is full-maze accuracy), our pilot-scale budget is far below the benchmark’s demands: at this paper’s standard budget both local arms score zero full-maze accuracy. Under a fair-shot scale-up (depth 48, hidden 64, 50 epochs, 20k mazes; 1 seed, exploratory), the predicted ordering emerges on the benchmark’s own metric: the 2D-local arm solves 11% of held-out 9×9 mazes with 0.97 pixel accuracy and fades to zero by 13×13 , while the matched 1D arm solves *none at any size*, its pixel accuracy stuck at the all-background floor (0.85–0.87). This is the procedure’s own step-5 warning made concrete: the task’s required depth scales with maze size, so a fixed-depth model cannot master it — mastery on this benchmark is achieved by test-time-extendable recurrence (Schwarzschild et al., 2021), consistent with our depth taxonomy (§6). The external check therefore confirms the *ordering* claim (2D-local $\gg$ matched 1D at every size, on a task we did not design) and the step-5 depth caveat; absolute benchmark mastery at pilot scale was not achieved, and we report it as such.

9 Limitations

- **Pilot scale.** Hidden width 32 and synthetic binary tasks throughout. The statistical base is hardened but small: all load-bearing comparisons were re-run at 5 seeds with per-seed records; the five headline effects are significant under Holm correction (all $p_{\text{Holm}} < 2 \times 10^{-5}$, one-sided paired t , $t \in [24.7, 53.5]$); per-seed W^* distributions are reported in §4 (both 1D arms at $W^* = 4$ in every seed); and every pre-registered gate re-checked unchanged on the 5-seed means (none flipped). Larger models and wider task batteries remain untested.
- **Task breadth.** The mechanism families are synthetic and designed to isolate bandwidth. Two out-of-sample checks on external benchmarks (§8.1: official CLRS-30 BFS; the Schwarzschild maze benchmark) now bound this concern but do not remove it; broader batteries remain future work, and the maze check in particular is 1-seed exploratory with pilot-scale absolute performance.
- **The capable-global comparison is partially closed.** After two training failures, a curriculum-trained Universal Transformer learns F1 in-distribution and fails to length-generalize (§7.1), so the boundary now rests on a trained model at pilot scale — with the depth-budget confound eliminated on the bounded-depth metric family (F4b), where the same trained model with ample iteration depth still decays to chance at unseen sizes. What remains untested is scale: whether much larger global models — cf. the scale-trumps-inductive-bias lesson of Dosovitskiy et al. (2021) — would close the transfer gap.
- **Supervised, hand-designed substrate.** All results concern what structure a learner can *exploit*, not what it can *discover*; the learned-layout null (§7.3) marks the current frontier of the latter, and our separate multi-year program on emergent 2D communication closed negative (14 pre-registered nulls), which we treat as out of scope here.
- **Two dimensions specifically.** We did not test 3D or higher; the bandwidth argument generalizes (bandwidth of a d -dimensional lattice under $(d-1)$ -dimensional layouts) and is an obvious extension.

10 Conclusion

Whether a 2D layout helps a neural network reason is not one question but three, and they have different answers. As an input encoding under global attention: no. As a representation to be learned from implicit structure: not yet — a set-transformer remains the wall. As a *computational medium* under a locality constraint: yes, decisively, and for stateable reasons: a bandwidth separation that no serialization can close, a scale-equivariance that turns small-instance training into large-instance competence, and a depth taxonomy that predicts when that competence is flat. In slogan form, locality gates dimensionality: 2D layout is a precise reasoning prior rather than a general one, and we provide the map of where it pays.

Reproducibility

All stages were pre-registered in-repo before execution, with raw per-seed results as JSON alongside; the statistical-hardening data (`harden_stats.json`) accompanies the code. Experiments run on a single laptop-class accelerator; every original pre-registered stage took under ~ 25 minutes, while the heaviest revision re-run (the 5-seed hardening of the learned-layout gate, executed under contention) took ~ 11 hours. Code, pre-registrations, and raw per-seed results are available at <https://github.com/sakatani/locality-gates-dimensionality>.

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